

do not care whether laborers lose their jobs and are not concerned about their living conditions; regarded from an extreme perspective, they also do not care if the laborer's family becomes homeless, maybe even dying or committing suicide. Capitalists want to do something about the downward process not because they feel sorry for laborers, but because their profits are decreasing and their equipment is idle. As for unemployment itself, capitalists are pleased because it drives down the money wage rate.

The fundamental classes of the capitalist society are the capitalist class and the wage labor class. They are mutually indispensable. Without wage laborers, capitalists cannot exist. Without capitalists, wage laborers cannot exist. Laborers can exist without capitalists, but as *wage* laborers, the existence of capitalists is a necessary condition. This is because one of the conditions for the existence of wage laborers is the purchase of their labor by capitalists.

What if capitalists are unable to purchase the labor from wage laborers? Laborers cannot make a living and thus suffer. However, it goes beyond this simple fact. When capitalists cannot purchase labor, the capitalist class loses the chain on which the capitalist class has bound the labor class. Capitalists are the ruling class because laborers have to be employed by capitalists to make a living. Hence, when capitalists cannot employ laborers, the labor class ought to seek ways to live without capitalists. This means that keeping a wage-labor system becomes difficult. Thus, the labor class, sooner or later, stands up for a fundamental reform of capitalist production relations. Cumulative mass unemployment shakes the foundation of the capitalist system.

If the downward disequilibrium cumulative process goes on without limit, it causes a deepening of disequilibria to the extent that capitalist reproduction itself becomes impossible. As long as the capitalist system lasts as a specific society for a historical phase, this downward cumulative disequilibrium process must be halted. The reversal from downward to upward is *inevitable*.

4.4.2 Triggers of Reversal

As revealed in the previous section, in a capitalist system, the suspension or the reversal of an upward disequilibrium cumulative process inevitably causes a downward disequilibrium cumulative process. Without this reversal of the disequilibrium cumulative process, sustainment or reproduction of a capitalist system would be impossible. Therefore, *as long as* we premise the persistence of a capitalist system, the downward disequilibrium cumulative process has to be reversed, and in this sense, reversal is *inevitable* as examined in Sect. 4.4.1.

In Sect. 4.4.2, we examine the triggers for reversal; please be reminded of the notes on triggers for an economic crisis, and that it is wrong to argue that reversal is inevitable due to a specific trigger based on the general characteristics of a capitalist system. What we can deduct from the general characteristics of a capitalist system is only a list of several triggers that seem to be highly probable (Sect. 4.3.2). We will not repeat the reasons here.

4.4.2.1 Resolution of Bottleneck

In the downward cumulative process, accumulation demand decreases at an accelerated speed; therefore, demand for production goods, labor, and funds also contract. Consequently, the difficulty of access to production goods, labor, and funds, i.e., the bottleneck that appeared in a favorable phase, is resolved. Excess production capacity, unemployment in the labor market, and idle funds in the financial market accrue. The resolution of these bottlenecks may trigger a reversal of capitalists' accumulation demand from decreasing to increasing. When the decrease of accumulation demand that sets off the downward process arises, being repressed by difficulties regarding access to production goods, labor, and funds despite capitalists' strong will for accumulation, the resolution of those bottlenecks may reverse the trend of accumulation demand from the repressed to the increasing condition again.

However, when bottlenecks and difficulties in access to production goods, labor, and funds are resolved, it does not *always* make capitalists' accumulation demand increase, failing to reverse the downward cumulative process. Even if production goods, labor, and funds become excess supply, turning their markets into a seller's market, capitalists might not increase accumulation demand. The opposite is true: it is more likely they may decrease even more. As the excess supplies of production goods, labor, and funds come from a production decrease due to demand and utilization rate decreases in the production goods and consumer goods sectors, the accumulation demand may decrease further. This may accelerate the downward disequilibrium cumulative process.

Others, however, argue that "excess supplies of production goods, labor, and funds cause a fall in the production goods price, money wage rate, and interest rate. As a result, the profit improves, capitalists increase the accumulation demand again, and the business cycle is reversed to an upward process." Suppose we apply the logic we already criticized in Sect. 4.2.2: shortages in the labor market lead to an increase in the real wage rate, which leads to a decrease in the exploitation rate and shrinking accumulation demand. Their logic can be summarized as follows: moderation in the labor market condition, i.e., an increase in unemployment, leads to a declining real wage rate, which leads to an increase in the exploitation rate and an increase in accumulation demand. Yet, as it is already apparent, this argument is incorrect. A decrease in the *money* wage rate does not equal a decrease in the *real* wage rate. In the downward process, commodity prices are falling, and as the result of the decrease in accumulation demand the real wage rate *increases*, leading to a *lower* exploitation rate.

Consequently, the profit rate decreases. In the case that commodity prices do not decrease, an even greater unsold stock and idle capacity accrue, so that the realized profit rate falls again. This makes accumulation demand contract even further.

4.4.2.2 Boom in Consumption Goods Sector

An essential feature of the downward disequilibrium cumulative process is the accelerating decrease in accumulation demand. However, as already clarified in Sect. 2.4.3 and mentioned in the argument regarding the triggers of economic crisis, when accumulation demand decreases (increases), the real wage rate in terms of production goods and consumption goods *always* increases (decreases). The amount of production goods production and the profit rate of the production goods sector always decrease (increase), but the amount of production and the profit rate of the consumer goods sector *can* increase (decrease). Although the real wage rate in terms of consumption goods increases while the exploitation rate decreases, the profit rate of the consumer goods sector can increase because the price of production goods can decrease more than the price of consumption goods. In the downward process, resulting from the decrease in accumulation demand, if the situation occurs as just explained, so that the amount of production, i.e., the utilization rate and the profit rate in the consumption goods sector increases, this can be one of the triggers for reversal.

More specifically, if the capitalists in the consumption goods sector increase accumulation demand in positive response to the increase in the utilization rate and the profit rate, and if this improves the capitalists' expectation regarding production goods sector, of which the utilization rate and the profit rate have decreased, it might increase the total amount of accumulation demand. This might halt and reverse the downward cumulative process.

However, it is wrong to think that it *necessarily* causes reversal. If the utilization rate and the profit rate increase in the consumer goods sector in the downward process, the capitalists in the consumption goods sector may not immediately decide to increase accumulation demand for the following reasons. In the downward process, capitalists' expectation becomes pessimistic. Even if the utilization rate and the profit rate show a temporary increase, they might not implement accumulation demand for enhancement of production capacity. Even if the capitalists in the consumption goods sector increase the accumulation demand, the capitalists in the production goods sector may strongly decrease the accumulation demand due to the decreasing utilization rate or profit rate. Therefore, the total amount of accumulation demand may continue to decrease.

4.4.2.3 Private Consumption of Capitalists

When the downward disequilibrium cumulative process proceeds and the accumulation demand continues to decrease, the private consumption of capitalists supports the demand. This may suspend or reverse the downward process because capitalists' demand for private consumption is different from accumulation demand. It is the demand that is less susceptible to the utilization rate and the profit rate, which fluctuates according to the business cycle. In this regard, it goes opposite to the laborers' demand for private consumption because the latter is susceptible to their

wage income and the business cycle. Therefore, in order to meet this constant demand for consumption by capitalists, a certain level of consumer goods production is necessary. On the other hand, when accumulation demand not only decreases but also falls to the negative level, replacement of exhausted production goods is not made so that the production capacity gradually decreases. Sooner or later, excess demand will arise in the consumption goods sector. Consequently, the capitalists in the consumption goods sector will order production goods. The production of the production goods sector will be revived, and this may cause an upward reversal. For that reason, capitalists' private consumption can be one of the triggers for reversal.

Capitalists' private consumption being held at some minimum level regardless of the business cycle phases, in the sense that the production level cannot fall below a certain level, will certainly play a role to *suspend* the downward process. However, we cannot assure that it can always cause a *reversal* from the downward to the upward process. That is because even if the capitalists in the consumption goods sector increase the demand for production goods, in case this is covered by a decrease of unsold stock in the production goods sector, it will not generate a production increase in the production goods sector. In addition, although this rise in demand may increase production in the production goods sector, it may not increase accumulation demand to a positive level.

4.4.2.4 Replacement Demand

Each fixed equipment has its own physical service life. *Physical* service life describes the period in which fixed equipment can work physically in the production process. So, fixed equipment can support the production capacity without replacement while they have service life. In this point, it is different from non-durable production goods such as raw materials that are exhausted and require replacement once they are used for productive consumption. When fixed equipment is replaced, it generates a sizable demand for the production goods sector. This replacement demand has the same effect as the accumulation demand on the current supply and demand relations. That is because the replacement demand for fixed equipment is greater than that for the part that is worn out by the current production.

If replacement demand for fixed equipment appears in a downward process, the market of the production goods sector improves directly while the consumption goods sector improves indirectly. This can be a trigger of reversal through changes, from a decreasing trend of accumulation demand to an increasing trend.

However, there is no guarantee that the replacement demand for fixed equipment necessarily becomes a trigger to reverse from a downward process to an upward process. First of all, even if some capitalists make replacement demands for fixed equipment, other capitalists may not follow and continue to make negative accumulation demands. As a result, the demand for production goods for the whole society may not increase. Secondly, if all capitalists make only replacement demands for fixed equipment, the demand for production goods increases only in the period when the replacement demand is implemented; other than during that period, demand for

fixed equipment remains low. This generates a so-called *replacement cycle*. However, this cycle does not have an upward trend. Although some people misunderstand this, the replacement cycle is different from the business cycle, and this cycle happens only at the bottom that is reached in the downward process of the business cycle.

Please note that the replacement demand for fixed equipment does not necessarily arise when *physical* service life comes to an end. It may arise earlier than its service life, and it may not arise even when it passes. That is because firstly, even if the *physical* service life is given—let us assume ten years, it is not necessarily profitable for capitalists to continue to use the fixed equipment for these ten years.¹⁸ In fact, in the downward process of a business cycle, as the real wage rate increases, the real wage rate may exceed the labor productivity under production with the fixed equipment. Then, even if the physical service life has, say, three more years left, they have to abandon this fixed equipment in seven years. It means that the *economic* service life becomes seven years. When it comes to the economic service life, replacement demand may arise earlier; not for the fixed equipment with the same technology as before, but for equipment embodied with higher labor productivity and new technology. Secondly, even though the physical service life (and thus the economic life) of fixed equipment has already passed, replacement demand may not arise. Capitalists may decide not to maintain the current production capacity, but to decrease it by creating negative accumulation demand. This is because they are pessimistic about the future due to a decrease in the utilization rate and profit rate in the downward process.

4.4.2.5 Introduction of New Production Methods

We examined the process that forces capitalists to switch to new technologies or to go bankrupt in Sect. 4.3.3.3. The gist of our argument was that, as the result of the increase in the real wage rate in a downward process, the force by which production methods with low labor productivity must be abandoned forcibly starts. Capitalists are forced to choose between bankruptcy or the introduction of new production methods. This force is the most *important* trigger that makes the capitalist system get out of the forced simple reproduction at low levels and turns upward. In order to introduce new production methods, capitalists have no choice but to purchase production equipment embodied with the new production methods. The amount of money that needs to be spent on this is far higher than to depreciate the current production. Thus, the demand for production goods, i.e., accumulation demand that exceeds the scale of replacement demand, is generated again.

When we considered how capitalists make decisions on accumulation demand in Sect. 4.1.3, we proposed that the capital accumulation rate, i.e., the proportion

¹⁸ Capitalists will not stop operating production equipment even when the sales price decreases below the cost, including the depreciation of production equipment. That is because it is profitable to operate as long as the sales price exceeds current expenses. Thus, they will not replace such production equipment in a situation like the one just mentioned.

of accumulation demand to existing capital, is an increasing function of the utilization rate or the profit rate. We argued that this plays an important role in giving a cumulative nature to the disequilibrium. This nature of decision-making on accumulation demand reflects the characteristic of capitalists' decision-making and, therefore, that of the capitalistic production relation. However, if all capitalists took this behavior in decision-making, under any circumstance, the downward cumulative process would not stop but increasingly progress, and it would finally stop only when it hits the floor that is supported by the capitalists' private consumption, as mentioned in Sect. 4.4.2.3. Moreover, it would not turn into the upward process unless there arose factors such as a large increase in expenditure by states and others for purposes other than reproduction. However, in the history of the business cycle that we know, a downward process has never hit the above-mentioned floor, and business cycles turned upward without increase in state expenditure, except for the period after the Great Depression in the 1930s.¹⁹ This means the following: On one hand, the way capitalists make decisions on accumulation demand has the nature which we described in Sect. 4.1.3. On the other hand, it also has such nature that capitalists increase the accumulation demand despite a low utilization rate and a low profit rate, possibly as low as even below zero. They do so precisely because of the low utilization rate and the low profit rate. Why do capitalists increase accumulation demand for the very reason that the utilization rate and profit rate are extremely low? It is because they are forced to go bankrupt, being not capitalists anymore unless they introduce new production methods and increase labor productivity. This enforcement is made by the increase in the real wage rate and is accompanied by the decrease in the exploitation rate in the downward process.

Of course, not every capitalist takes such actions of forced introduction of new technology and increasing accumulation demand. Firstly, those capitalists who have already adopted production technology with higher productivity, mainly large companies, may feel less pressure from such enforcement than other capitalists, even in the later phase of the downward process. Secondly, the capitalists with less fundraising ability are unable to get funds to purchase the equipment for the introduction of new technologies, even when they are forced to introduce new technologies. They have no choice but bankruptcy. In any way, since some capitalists increase accumulation demand to escape bankruptcy by responding to the enforcement in the downward process, the capitalist system can reverse sooner or later and continue to reproduce itself without depending on "external" factors such as state intervention.

As expressed above, an increase in accumulation demand for the introduction of new production methods is one of the most important triggers to reverse the downward process. However, this does not necessarily mean that a capitalist system *immediately* begins an upward process. In this point, it is different from the case in which a downward process begins as soon as accumulation demand decreases. Under slightly improved market condition due to the emergence of accumulation

¹⁹ At the time of crisis 1929, the capitalist system was already at the stage in which relying on "self-recovery" put the system on the verge of a crisis; that is to say, a general crisis. For this reason, state monopoly capitalism was required and was established.